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Structure of Pelagic Food Chain and Relationship Between Plankton and Fish Production[1]

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Further observations on the standing stocks of pelagic organisms confirm the occurrence of approximately equal biomass over logarithmically equal size ranges. A simple theoretical framework is developed that shows that the structural elements of the pelagic ecosystem can be described in terms of the sizes of predator and prey and of the efficiencies of their interactions. In practice this means that if the standing stock at any size range is known, the standing stock at any other size can be estimated, and if the growth rate at this size is known, the production can be estimated. The theory is tested on three fisheries. For the Gulf of Maine and the North Sea, phytoplankton production is estimated from fishery production. For the area off Peru the fishery production is estimated from the plankton production.

Key words: pelagic ecosystem, predator–prey relationships, plankton production, marine fisheries, Peru, North Sea, Gulf of Maine

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Si l'on ordonne les organismes pélagiques selon leur taille sur une échelle logarithmique, on trouve que la quantité de matière vivante est égale pour chaque taille. C'est-à-dire que la concentration (milligrammes par litre) soit de plancton, de poissons ou d'organismes intermédiaires est la même dans chaque intervalle logarithmique. Nous avons formulé une description mathématique simple qui démontre que l'on peut décrire la structure de la chaîne alimentaire pélagique en ne considérant que la taille des prédateurs et de leurs proies, et les rapports existant entre les deux. En utilisant notre formulation on peut estimer la production d'une pêcherie en mesurant par exemple, la quantité de plancton et le taux de croissance des poissons. Nous avons appliqué la théorie à trois pêcheries. Dans le cas du golfe du Maine et de la mer du Nord, nous avons estimé la production du plancton par rapport à la production des poissons. Pour la région côtière du Pérou nous avons estimé la production des anchois par rapport à la production du plancton.

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OCEANIC pelagic environments are remote from normal research facilities. They are difficult to get to and even more difficult to stay in, and this makes the pelagic ecosystem somewhat less than an ideal choice for study. This is further compounded by the fact that, at least in the subtropics, the ecosystem structure seems to be incredibly complex. The number of species is large and their apparent web of interactions is, at first sight, bewildering. However, there are certain features of the pelagic environment that are appealingly simple. Climatic and ecological stability are obvious examples. The basic trophic structure also seems to be simple (Sheldon et al. 1972; Kerr 1974) even though the details are undoubtedly complex. In this study we extend the simplified view of the pelagic food chain, based essentially on particle size spectra, and show that by considering the interactions of organisms relative to their size we can arrive at a plausible structure for the pelagic food chain; from this we derive reasonable estimates of plankton and fish production.

Structure of Pelagic Food Chain

PRACTICAL CONSIDERATIONS

We can characterize the trophic structure of pelagic ecosystems by considering three factors (Sheldon et al. 1973). The first concerns the nature of the primary production. With some notable excep-

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tions (*Sargassum* for instance), most of the primary organic production in the sea is by single-celled plants. The second concerns the nature of the first trophic link. Most herbivores are small, but they are usually larger than the plants. The third concerns the relationships of the higher trophic levels. In the sea, organisms at higher trophic levels tend to be larger than those at lower trophic levels. It is, in fact, virtually a biological law that among the plankton and the nekton a predator is larger than its prey (Marshall 1971, p. 6), although this is not so generally true of the benthos. It is easy to see why this should be. An aquatic animal lives in a medium 800 times denser than air and efficient movement can only be achieved by streamlining (Marshall 1971, p. 23). Appendages that could be used for grasping a large food object are inconvenient and are not commonly developed. A predator must take its prey by swallowing it and must therefore be larger than the prey.

The structural limits imposed by a truly aquatic mode of life allow relatively little variation in form, and in contrast to, say, terrestrial animals, most pelagic fish have a similar shape. As the shapes are similar one would expect that the ratio of size of predator and prey (at least at the higher trophic levels) would also be fairly constant.

In a steady state system a predator can only consume the growth of its prey population. It must not make permanent inroads into the basic stock. In other words, it must live on the interest and not touch the capital. As the growth rate of a population will vary with the average size of the organisms (Fenchel 1974), the stock of prey required to provide a predator with a given amount of food will also vary with the size of the prey. A small stock of small prey could provide the same production as a large stock of large prey. In terrestrial systems a predator of any particular size could, theoretically, have a wide choice of prey size. A 25-kg predator could, for instance, live on ants, rabbits, or antelopes (depending on its own evolutionary adaptations) and the standing stock of its prey would vary accordingly. But in the sea there are fewer options available. Whatever its adaptation or mode of life a predator tends not to live on prey that is very much larger or smaller than about 7% of its own length (D. Ware personal communication). Those that do, e.g. lampreys, tend to follow unusual and specialized adaptations and are beyond the scope of our discussion. One would expect, therefore, that as the size ratio of predator and prey does not vary greatly, then the ratio of the standing stocks will also be similar.

Some observations have been made that suggest that standing stock does not vary much with size (Sheldon et al. 1972), and recent theoretical studies indicate that given the observed predator to prey body size ratios, the ratios of standing stocks may approximate to unity (Kerr 1974). As the predator and prey populations are likely to each have the same logarithmic variation in size, it follows that standing stocks over logarithmically equal size intervals will be similar (see Sheldon et al. 1972 and Fig. 1).

The original concept of roughly equal standing stocks within equal logarithmic size intervals was developed mainly from observations in the size range 1–100 μm, with a few observations at sizes around 1 mm (Sheldon et al. 1972). We have now made further observations in the Sargasso Sea that extend over a size range from 0.5 μm to about 2 cm (Fig. 1). This covers more than 4 orders of magnitude of linear dimensions. As the complete food chain from primary producer to terminal predator extends over <6 orders of magnitude, the observations cover more than two thirds of the food chain in terms of size and also account for two thirds of the total biomass.

Particle size distributions shown in Fig. 1 were derived from two sets of samples. Small particles (0.5–80 μm) were collected with a float and bottle sampler (Sheldon et al. 1973) at a depth of 4 m, and the particle size distributions were analyzed with a Coulter counter (Sheldon and Parsons 1967a). The adenosine triphosphate (ATP) content of prescreened (160 μm) samples was determined by the method of Holm-Hansen and Booth (1966). Large particles (zooplankton >250 μm) were collected

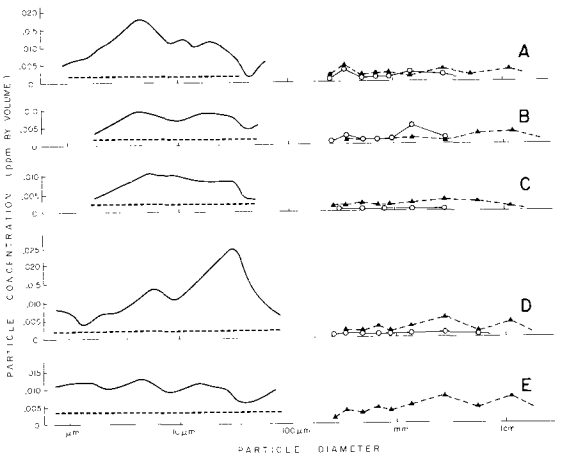


FIG. 1. Pelagic standing stocks (concentration per size grade) in the surface waters of the Sargasso Sea. O——O, Zooplankton concentrations, day samples; Δ——Δ, zooplankton concentrations, night samples; ———, total concentration of small particles; ------, average concentration of small living particles estimated from ATP measurements. A, 25°54′ N, 62°45′ W, February 6, 1975; B, 25°56′ N, 62°45′ W, February 7, 1975; C, 25°33′ N, 62°40′ W, February 8, 1975; D, 32°51′ N, 62°41′ W, February 11, 1975; E, 32°45′ N, 62°32′ W, February 13, 1975.

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with a conventional ($\frac{3}{4}$ m diam) metered net towed across the wind at 5 m depth. Size distribution of the zooplankton was determined by wet sieving the fresh material immediately after sampling. Sieves of various sizes were used, depending on the sample, so that measurable amounts of zooplankton were retained on each. Results have been recast in Fig. 1 to show the volume of material in a standard size grade. Each grade covers a relative diameter range of $2^{\frac{1}{2}}$. This scale has been fairly widely used in studies of marine particle and plankton distributions (e.g. Sheldon and Parsons 1967b; LeBrasseur et al. 1969; Parsons 1969; Sheldon et al. 1972; Poulet 1974; Conover 1977; Sheldon 1977). It is similar to the standard scales used for granulometric analysis (Krumbein and Pettijohn 1938; Sheldon 1969).

It is easy to see whether or not the individuals in a freshly taken zooplankton sample are alive, and the right-hand side of Fig. 1 represents the amount of living organisms (per standard size grade) in the size range from about 250 μm to 2 cm. However, the situation with small particles is less simple. We were able to see living organisms and nonliving particles in filter concentrates but we did not know in what proportion these occurred in the original samples. For an estimate of living material we had to depend on measurements of ATP content of the total sample. Solid curves on the left-hand side of Fig. 1 represent the particle concentration per size grade, as measured, and the broken lines represent the average concentration of living material per size grade as estimated from the total ATP values.

Figure 1 shows that the hypothesis tentatively advanced by Sheldon et al. (1972) — that roughly equal standing stocks occur over logarithmically equal size intervals — is confirmed within the limits of accuracy of our observational techniques. In the next section we develop a structure for the pelagic food chain based on the assumption that the logarithmic distribution of biomass is constant and independent of size. We then extend this to consider the range of biomass distribution that could reasonably occur.

Theoretical Considerations

If we define production as the formation of living material, then the relationship between the production of a predator and its prey can be expressed as

$$P_c = P_p \cdot G_e \cdot C_e \tag{1}$$

where P_c is the amount of predator tissue formed, P_p is the amount of prey tissue formed, G_e is the growth efficiency (defined as the quantity of predator tissue formed relative to the amount of prey ingested), and C_e is the predation efficiency, i.e. the proportion of the prey production taken by the predator. Equation (1) is not particularly useful because none of the quantities can be measured easily, but we can substitute into the equation variables that we can measure, and from these we can estimate those variables that we cannot measure. In a steady state system, natural populations will tend to grow exponentially and this growth will be removed as fast as it is formed. Under such conditions the production is expressed simply as $P = r\,S$, where r is the instantaneous growth rate and S is the standing stock. In any one population the average growth rate is constant, but it will vary between populations depending on the average size of the organisms (Fenchel 1974).

We can therefore write (1) as

$$r_c S_c = r_p S_p \cdot G_e \cdot C_e \tag{2}$$

where r_c, S_c, and r_p, S_p are the instantaneous growth rates and standing stocks of predator and prey, respectively. Rewriting (2):

$$r_c/r_p = S_p/S_c \cdot G_e \cdot C_e \tag{3}$$

If S_p/S_c is approximately unity, it follows that

$$r_c/r_p = G_e \cdot C_e \tag{4}$$

There is a definite relationship between growth rate and body size that is simply a variant of the well known "metabolic law." This can be expressed as $r = aW^n$, where W is the body weight and a and n are constants. Fenchel (1974) considers n to have a value of around -0.275. We have also calculated n (but using only data for plankton and fish) and arrive at the expression $r = aW^{-0.24}$, or $r = bD^{-0.72}$, where D is the equivalent spherical diameter and a and b are constants that depend on the environmental temperature. The ratio of growth rates can therefore be expressed as a ratio of sizes.

$$r_c/r_p = (D_c/D_p)^{-0.72} = G_e \cdot C_e \tag{5}$$

For the purposes of this study, equation (5) describes the size structure of a pelagic food chain with equal standing stocks of predator and prey.

This size structure is not subject to much variation. Copepods will take prey over a wide range of size (Conover 1977) with an average ratio in the range 20–30 (Sheldon et al. 1972; Poulet 1974), and a ratio of around 100 could be reached by certain oceanic pteropods (Gilmer 1972) or tunicates (Alldredge 1972) that feed directly on nanoplankton. But for many pelagic organisms the size ratio of predator to prey tends to cluster around a value of 14 (Fig. 2). If we substitute this average value in equation (5) we find that $G_e \cdot C_e = 0.15$; that is, if a predator takes all the prey production, then it must have a growth efficiency of 15%. If only 80% of the prey production is taken, then the growth efficiency must be 19%. If larger prey are taken (i.e. $D_c/D_p < 14$) then the growth efficiency must increase in spite of the fact that all the prey production is taken. This might be difficult for a predator to achieve. In

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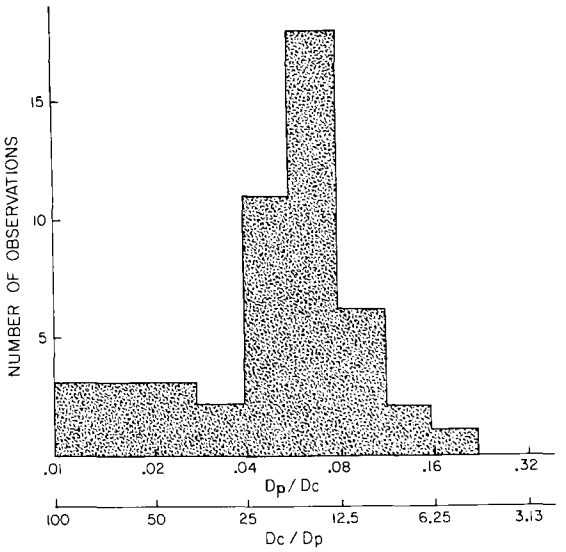


FIG. 2. Histogram of the ratio of the size of various pelagic predators relative to their prey. D_C = equivalent spherical diameter of predator and D_p that of prey. Data sources as follows: copepods: Poulet (1974), Parsons et al. (1969), Parsons and LeBrasseur (1970); herring: Blaxter (1973) Lebour (1924) Guryeva (1973); young salmon: Parsons and LeBrasseur (1970); redfish larvae, Cod larvae: Bainbridge and McKay (1968); haddock larvae, Pollack larvae: Marak (1960).

evolutionary terms it might be easier for a predator to adapt to smaller prey. With D_c/D_p at 20, equation (5) would be $G_e \cdot C_e = 0.11$. This means that with 15% growth efficiency a predator would only have to take 73% of the prey production. However, a trend towards utilizing smaller prey could be self-defeating, as small prey may be difficult to catch in quantity. This difficulty would be relieved somewhat if the prey occurred in patches.

The structure of the pelagic food web might not be as restrictive as equation (5) indicates. In equation (3) we eliminated S_p/S_c by assuming that it was unity. While this is probably true on average, the ratio varies with the particular predator–prey relationship considered. In Fig. 3 we have rearranged equation (3) and plotted the relationship between the ratio of the standing stocks and the ratio of the sizes of predator and prey. The three lines are for different values of $G_e \cdot C_e$. If we assume that C_e is unity, i.e. all the prey production is taken by the predator, then the lines represent the relationships for growth efficiencies of 10, 15, and 20%. If we take an average predator to prey size ratio of 14 (Fig. 2) and project it to the average growth efficiency of 15%, the corresponding ratio of standing stocks is close to 1. This we assumed in development of equation (5). If we consider a range of predator to prey ratios from 5 to 25 (Fig. 2) and consider a range of growth efficiencies from 10 to 20%, then the standing stocks could vary from about 0.3 to 2.0. The hatched area represents the range of structure that could reasonably occur, although the range in any one system would probably be less than this.

Relationship Between Plankton and Fish Production

The trophic structure we have postulated describes the interaction of predator and prey, and to relate fish and plankton production one would have to know the number of predator–prey links involved. However, we can eliminate the consideration of trophic levels and relate plankton and fish directly if we make certain reasonable assumptions.

1) Predator efficiency is 100%. According to the review by Steele (1974) the plant production in the sea is rapidly used by the herbivores, and there is little reason to doubt that a similar situation exists at higher trophic levels. However, it is very difficult to demonstrate unequivocally that 100% predation is the rule. McAllister (1969) gives data for monthly mean standing stocks of phytoplankton and zooplankton for 6 yr at Ocean Station P (50° N 145° W) in the North Pacific (McAllister 1969, fig. 3). We recast his data and obtained the following relationship (in terms of milligrams carbon per square metre):

$$\text{Zooplankton} = 1.5\ (\pm\ 0.3)\ \text{Phytoplankton} - 270\ (\pm\ 180)$$

The ratio of 1.5 would, according to McAllister (1972, fig. 1), indicate a transfer efficiency (equivalent to growth efficiency if predation = 100%) of 15%. If the size ratio of the zooplankton to the phytoplankton averaged around 30 (Sheldon et al. 1972, fig. 13), we would also predict a growth effi-

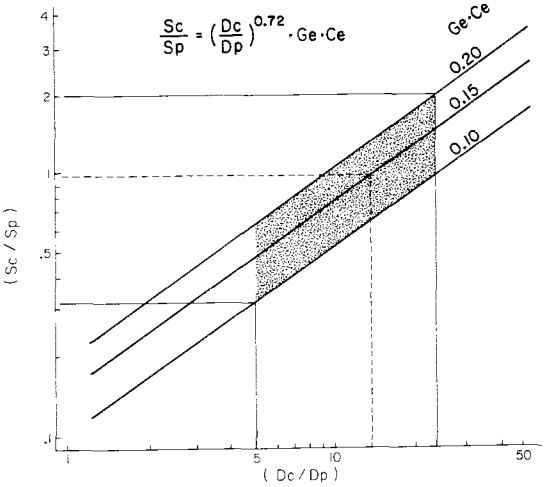


FIG. 3. Relationship between standing stock, size, and growth efficiency of predator and prey (for further explanation see text).

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ciency of 15% (see Fig. 3), providing that the predation was 100%.

2) Standing stocks are similar at all sizes. This is not a particularly new concept. Harvey (1950) showed that in the English Channel the standing stocks at different trophic levels were roughly similar, and Odum (1971, fig. 3–6), working from energetic considerations, has shown very clearly why marine and terrestrial standing stock patterns are fundamentally different and why marine standing stocks are likely to be roughly constant and independent of the size of the organisms concerned. It follows from this that the production at any trophic level depends only on the growth rate, and thus the size of the organisms concerned (equation 5).

3) Apart from seasonal variation, most ecological systems are in a steady state. Many studies have been made since the idea of the "balance of nature" was first formally proposed (Darwin 1859), and it is now considered to be self-evident that in the long term, ecological systems must be in a steady state.

If these assumptions are accepted, then the production at any trophic level can be calculated if the standing stock or production at one trophic level is known. To illustrate this we consider two examples. In the first we start with a known fish production and estimate the plankton production needed for this, and in the second we start with a known plankton production and estimate the maximum fish production that could occur.

Gulf of Maine Fishery and the Plankton Production Supporting It

The production of individual fisheries in the Gulf of Maine varies greatly from year to year, but until the large foreign fleets appeared in the 1960s the total catch for all fisheries had remained more or less constant for many years (Sutcliffe et al. 1977). It seems that (during 1940–59) as the production of one fishery declined fishermen switched their efforts to another, and in this way they extracted a constant yield from the environment.

To estimate the plankton production needed to support the fish we assume initially that the total fish landings represent the total fish production. This is incorrect, but it is used simply to provide a base for calculation and will subsequently be modified. We further assume that the average size of the fish is similar to that of a small Atlantic cod (*Gadus morhua*) and that the relative size range of the fish is about half that of the phytoplankton; i.e. the smallest and largest fish differ by a factor of 4 in linear dimensions, while the phytoplankton differ by a factor of 10. Total standing stock of the fish would then be about 70% of the standing stock of the phytoplankton (Fig. 4). If these assumptions are accepted, it is only necessary to know the average growth rates of the plankton and fish (plus the total fish catch) for an estimate of annual phytoplankton production to be made.

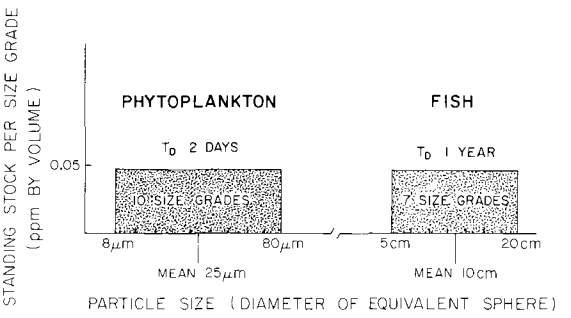


FIG. 4. Relationship between size, standing stock, and growth rate for phytoplankton and fish in the Gulf of Maine. (Note that for sizes to be compared a common scale is used — the diameter of an equivalent sphere. Actual fish lengths would be 3–4 times greater, depending on shape.)

In the Gulf of Maine, fish the size of cod could possibly double their standing stock in about a year (Sheldon et al. 1972, fig. 13). Annual catch should therefore equal the standing stock, which would be the same as the annual production, bearing in mind the assumption made above. As the area of the Gulf of Maine (to the edge of the continental shelf) is about 1.4×10^{11} m^2 and the average annual fish catch is about 4.2×10^{11} g (Fig. 6), the average annual fish production works out to be 3.0 g·m^{-2}·yr^{-1} (wet wt).

Average growth rate of the phytoplankton in the Gulf of Maine is not easy to estimate because there is much seasonal variation. The maximum rates that have been recorded elsewhere indicate a possible minimum doubling time for the Gulf of Maine phytoplankton of about a day (Cushing 1959; Eppley 1972; Sheldon et al. 1972), but the average doubling time is probably nearer to 2 days (Eppley and Strickland 1968).

Production rate of the phytoplankton should therefore be about 180 times greater than production rate of the fish. After taking into consideration the differences in standing stock, total phytoplankton production should work out to about 260 times the total fish production. This would be about 780 g·m^{-2}·yr^{-1} (wet wt). In terms of organic carbon this represents about 45 g C·m^{-2}·yr^{-1} (C/V ratio from Mullin et al. 1966). The available estimates for primary production on the northeastern seaboard of North America would lead one to expect a production in the Gulf of Maine of about 100–200 g C·m^{-2}·yr^{-1}. Our estimate is therefore rather more than a factor of 2 too low.

The reason for this lies with our original assumption that fish catch represented all fish production. The relationship between fish landed from a fishery and actual fish production is not completely understood, but reasonable limits can be set. Dickie (1972)

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has suggested that for the maximum sustained yield the catch can be no more than about half the production. This would give us a minimum estimate of plankton production of 90 g $C \cdot m^{-2} \cdot yr^{-1}$. On the other hand, Edwards (1968) has estimated that the annual fish catch in the Gulf of Maine represents about one quarter of the standing stock. This would give a possible maximum estimate for phytoplankton production of 180 g $C \cdot m^{-2} \cdot yr^{-1}$. We should also consider that fish production will occur in species not taken commercially, but there would have to be a significant production by noncommercial fish of the same size as the commercial species for this to affect our estimates.

An obvious criticism of these estimates is that because we knew the answers and because we had to make certain assumptions, it was not too difficult to arrange the assumptions to give the correct answers. We can partially satisfy such criticism by applying exactly the same assumptions we used for the Gulf of Maine to the fisheries of the North Sea. Steele (1974) gives data on the North Sea fish catch that indicate that between 1909 and about 1955 the total fish landings remained fairly constant at about 1×10^6 $t \cdot yr^{-1}$. Using the data given by Steele (1974) and using exactly the same assumptions as we have outlined above for the Gulf of Maine, we obtain initial estimates of fish production of 2 g $\cdot m^{-2} \cdot yr^{-1}$, and plankton production of around 500 g $\cdot m^{-2} yr^{-1}$ (wet wt), which is equivalent to about 30 g $C \cdot m^{-2} \cdot yr^{-1}$. If, as in the Gulf of Maine, we assume that steady state landings represent between a quarter and a half of the fish production, we would estimate North Sea phytoplankton production to lie between 60 and 120 g $C \cdot m^{-2} \cdot yr^{-1}$. Primary production in the North Sea is around 70–90 g $C \cdot m^{-2} \cdot yr^{-1}$. (Steele 1974).

We have related the estimates of fish and plankton production according to our ideas on the structure of pelagic food chains in spite of the fact that the fisheries in the Gulf of Maine and the North Sea are pelagic and demersal. However, providing that we consider only the primary and terminal producers our estimates are valid.

A proportion of the phytoplankton production will settle from the surface waters (Stephens et al. 1967; Hargrave 1975) and therefore will cease to be part of the pelagic system. But this production is consumed by benthic organisms that are themselves consumed by demersal fish. In this way all the phytoplankton production is ultimately funneled by either the pelagic or demersal route into fish production. For our purposes this food chain can be considered "pelagic." We could not have made similar estimates to relate phytoplankton production to zooplankton production, for to do this we would have had to have known the proportion of the phytoplankton production that remained in the surface waters and was thus available to the zooplankton.

Plankton Standing Stock off Peru and the Anchoveta Fishery Supported by It

It is little more than an interesting academic exercise to use data from a heavily exploited fishery, collected over many years, simply to estimate phytoplankton production. It is much easier to measure plankton production directly. However, the reverse process is possible. If the phytoplankton production or standing stock is known, the fish production can be estimated.

The anchoveta fishery of Peru differs from the fisheries of the Gulf of Maine and the North Sea in that it is truly pelagic. High levels of utilization, particularly of primary production, are typical of pelagic food chains (Greze 1970), and very little settles out (Riley 1970; Ryther et al. 1971). With such a system we are not limited to considering only primary producers and terminal predators. We can estimate the standing stock or production of any trophic level from a knowledge of any other. In the following treatment we estimate the anchoveta production from a knowledge of phytoplankton and zooplankton standing stocks and anchoveta growth rate.

Phytoplankton standing stocks of the Peruvian region are not well known but primary production has been measured. Strickland et al. (1969) found that production varied from 0.25 to 4.1 g $C \cdot m^{-2} \cdot day^{-1}$, with most of the observations being greater than 1 g $C \cdot m^{-2} \cdot day^{-1}$. Barber et al. (1971) measured production of 1.1 and 3.5 g $C \cdot m^{-2} \cdot day^{-1}$ at two stations, and Cushing (1971) considered a range of between 1 and 4 g $C \cdot m^{-2} \cdot day^{-1}$ to be typical of an upwelling area. Ryther et al. (1971) found that production varied from about 1 to 11 g $C \cdot m^{-2} \cdot day^{-1}$, but was typically around 5 g $C \cdot m^{-2} \cdot day^{-1}$. The average situation given by Cowles et al. (1977) shows a band of high production (5 g $C \cdot m^{-2} \cdot day^{-1}$) within 95 km of the coast and an area of lower production (in the range 1–5 g $C \cdot m^{-2} \cdot day^{-1}$) extending about 300 km offshore. A production in the range 2–4 g $C \cdot m^{-2} \cdot day^{-1}$ would indicate a standing stock of about 2–4 g $C \cdot m^{-2}$, as the doubling time of the phytoplankton is about a day (Strickland et al. 1969). This would be equivalent to about 35–70 g $\cdot m^{-2}$ wet wt (assuming an average size of 25 μm); but to estimate the standing stock per size grade we have to know the size range of the phytoplankton. In the absence of better data we assume as before that they occupy 10 grades. The phytoplankton standing stock will therefore be in the range 3.5–7.0 g $\cdot m^{-2}$ per size grade.

According to Reid (1962) the average zooplankton standing stocks off Peru vary from 30 to 60 g $\cdot m^{-2}$. Net-caught zooplankton typically occupy 7 size grades (Sheldon and Paranjape unpublished data). Average concentration would therefore fall in the range 4–8 g $\cdot m^{-2}$ per size grade. Using data of Pavlov

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(1968) for a similar upwelling region off West Africa we obtain a zooplankton standing stock of 5 $g \cdot m^{-2}$ per size grade.

We now have to consider how the level of anchoveta standing stock relates to that of phytoplankton and zooplankton. In the examples for the Gulf of Maine and the North Sea we were dealing with broad averages. We considered all kinds of catchable fish, pelagic and demersal, and the production by these would depend on many kinds of interactions at all stages of the food chain. We therefore assume for these cases that the average conditions indicated in Fig. 4 would apply and that plankton and fish standing stocks would be similar.

The food chain leading to the anchoveta is much simpler. It consists ideally of diatoms, copepods, and anchoveta, but some of the anchoveta bypass the copepods and take diatoms as part of their diet (deMendiola (1971). As we know roughly the relative sizes of the three groups making up the food chain, we can attempt to demonstrate the relationships between the standing stocks of the anchoveta and those of the smaller organisms of the food chain.

In the first link of the food chain the copepod–diatom size ratio is likely to be around 30 (Seldon et al. 1972; Poulet 1974). We would expect from this (see Fig. 3) that the standing stock of the copepods would be 1.5 times that of the diatoms (Fig. 5). On the other hand, the anchoveta–copepod size ratio is around 15 and we would expect the standing stocks to be equal; but as some anchoveta feed on diatoms this should lead to a relative increase in the anchoveta standing stock levels. We assume therefore that the anchoveta stocks are similar to the upper limit of the zooplankton stocks (Fig. 5) and that the anchoveta has an average standing stock of 8 $g \cdot m^{-2}$ per size grade. According to the growth data of Jordan (1974), the smallest and largest individuals (in the fishery) will not differ by more than a factor of 8 (by wt) and will occupy 3 size grades. The total standing stock of anchoveta in the fishery will therefore be 24 $g \cdot m^{-2}$.

The area of upwelling off Peru is about 5×10^{11} m^2

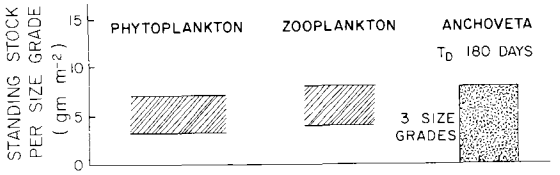


FIG. 5. Relationship of standing stocks of phytoplankton, zooplankton, and anchoveta off Peru. Hatched areas show the calculated range of phytoplankton standing stock and the observed range of zooplankton standing stock. (For further explanation see text.) Note that only the standing stock per size grade of the plankton need be known (not their size range) and that only the size range and growth rate of the anchoveta need be known (not their absolute size).

(Longhurst 1971), and according to FAO data (Anon. 1972), anchoveta are found (up to 300 km offshore) over a similar area. But according to Idyll (1973), the anchoveta fishery extends only 100 km offshore in an area corresponding to the zone of high average primary production defined by Cowles et al. (1977). We have taken Idyll's distribution as probably being the most correct representation and arrive at 2.4×10^{11} m^2 for the area of the fishery. Total anchoveta standing stock should therefore be about 60×10^{11} g.

From the growth pattern of anchoveta (Jordan 1974) one can estimate that a population growing exponentially would have an instantaneous growth rate of about 0.00385 $\cdot day^{-1}$. Such a population would double its biomass in about 6 mo. Yearly production will depend on how often the growth is removed from the population. If a day's growth is removed each day, then the yearly production would be 85×10^{11} g. If, on the other hand, the population was allowed to double before the growth was removed, then the production for the year would simply be twice the standing stock, or 120×10^{11} g. Clearly neither of these two extremes is probable and the true yearly production (available to a fishery) is probably somewhere between the two estimates. An estimate of annual production of, say, 100×10^{11} g would seem reasonable. Actual landed catch ranged from 85 to 123×10^{11} $g \cdot yr^{-1}$ between 1966 and 1971. (Fig. 6)

Discussion

When predictions from theory agree reasonably well with observation it does not necessarily mean that the two are perfectly in tune. The question of accuracy has to be considered, and this is not a particularly easy question to answer. However, it is possible to give some indication of the accuracy of our predictions.

Fisheries of the Gulf of Maine and the North Sea are the easiest to deal with. In these areas fish catch and area of the fishery are well documented. The food web is complex, which means that in spite of variations in predator and prey interactions many of the variations of standing stock are smoothed out before the end of the food chain is reached. The movement of fishing effort from one species to another also smooths out variations in total production. This means the average situation shown in Fig. 3 tends to apply, and it is possible therefore to predict fish standing stocks fairly accurately from average phytoplankton standing stocks.

For a sustained fish yield in a temperate region the annual fish catch should be between one quarter and one half of the standing stock. (Edwards 1968; Dickie 1972). Growth rate of the fish is such that the doubling time (T_d) of many commercial species is about a year (Sheldon et al. 1972). In this case fish catch would also be between one quarter and one

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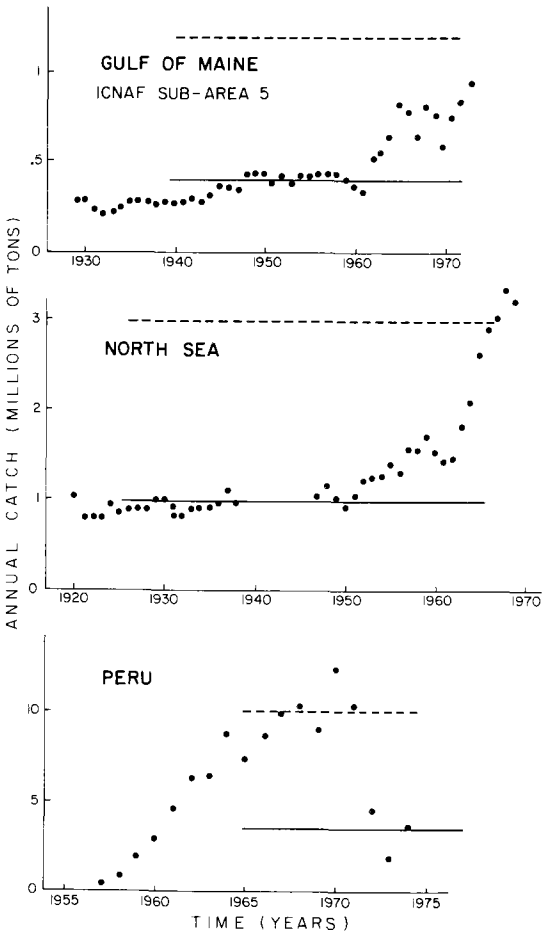


FIG. 6. Annual landings of all fish (pelagic and demersal) from the Gulf of Maine and the North Sea and of anchoveta from Peru. ———, Predicted fish catch if one third of the natural fish production is taken; ------, predicted fish catch if all the natural fish production is taken.

half of fish production. In Fig. 6 we show the fishery production we would predict (from considerations of phytoplankton standing stocks) if one third of the natural production were taken by the fisheries. This fits very well for those times when the same amount of fish was taken year after year; and this is a good indication that a sustainable yield was exploited. Whether or not this was the maximum sustained yield is, of course, open to question. If the maximum sustained yield is about one half of the natural fish production (Dickie 1972); we would predict that the Gulf of Maine should produce 0.6×10^6 t·yr^{-1} and the North Sea 1.5×10^6 t·yr^{-1}. Present production in the Gulf of Maine is at about this level but the level for the North Sea was exceeded in 1962.

While the food web of the Peru system is simpler than those of the North Sea and the Gulf of Maine, prediction is more difficult. This occurs because with fewer links and interactions there is less chance of stabilization by the averaging effect of many variables. However, the predicted fishery yield, on the assumption that one third of the natural fish production is taken by the fishery, gives estimates that are not greatly different from the recent catches. But there is considerable doubt as to whether the present catch levels represent a stabilization of the fishery after the boom years of the late 1960s or whether it is simply a reaction to the *El Niño* phenomenon.

In the case of the anchoveta the maximum sustained yield is not easy to define. We are not dealing here with a relatively slow-growing terminal predator; rather more than one half of the natural fish production could possibly be taken by a fishery without unbalancing the system. After the 1965 *El Niño* the fishery increased but the seabird population did not; yet after the 1957 *El Niño* the seabird population increased rapidly (Schaefer 1970; Idyll 1973). At that time there was no significant fishery in the area. From this we could conclude that the "natural" and fishing mortalities are related. As fish catch increases the amount available to the seabirds decreases. In recent years the birds are believed to have taken about 25% and the fishery 75% of the anchoveta production (Idyll 1973). If anchoveta mortality were due only to birds or fishing, then we could say that the maximum sustainable yield, predicted from zooplankton standing stocks, would be around 7.5×10^6 t·yr^{-1}. This is very close to the estimate of 8×10^6 t·yr^{-1} (quoted by Longhurst 1971), which was derived from catch effort data.

The major uncertainty in predicting the Peru landings lies in the estimate of the area of the fishery. In the North Sea and the Gulf of Maine good records are kept of where the fishing boats operate and hence the area of the fishery is known with reasonable accuracy. This is not so off Peru. Fish landings are well documented but the area from which the fish are taken is not. Therefore, even if prediction of fish production per unit area were perfectly accurate, the prediction of the total fish production would only be as accurate as the estimate of the total area of the fishery.

The advantage, as we see it, of using the food chain model we have developed to attempt to predict fish production is that it can be done in an area of unknown potential before a commercial fishery is started. In contrast, the conventional methods of assessing fish production — stock assessment or catch-per-unit-effort data — require that a fishery, or at least biological sampling that approximates to a fishery, be in operation at the time that the estimates are made.

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